Spacecraft Quaternion Based Attitude Input-Output Feedback Linearization Control Using Reaction Wheels

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***Abstract*—The aim of this paper is attitude control of spacecraft utilizing nonlinear control feedback linearization. Input-output linearization is a technique of feedback linearization theory. In this method choosing of output functions is very important and has a direct effect on stability. Here state variables are selected as output functions. In order to make sure that internal dynamic is stable and performance of controller is acceptable, zero dynamics analysis has handled. Optimal torque of each reaction wheel is determined based on distribution matrix of configuration and torque magnitude optimization of actuators. The simulation results show that the method has been successful to control the attitude of spacecraft.**

Keywords—nonlinear control; quaternion; zero dynamic; input-output linearization

I. INTRODUCTION

Space programs are an attractive area [1-4] and some new missions as formation flying [5-7], satellite-debris collisions [8] and spacecraft orbital maneuverer [9-10] interested for researchers. The attitude control is one of the most important subsystem of spacecraft. Many studies have been done on this filed including mathematical modeling [11-13], attitude control [14, 15] and control of attitude and orbital coupling [16]. Also attitude control design is need for some other applications such as quadrotor attitude control [17-18].

The problem of attitude control of spacecraft has received a great deal of attention in recent years. Dynamic of spacecraft equations are nonlinear hence nonlinear control design is often required. Nonlinear control theories usually are complicated. Feedback linearization is a nonlinear control approach that in recent years has been used in various research fields [19, 20]. For example, in [21], an input-output linearization design has been used for trajectory tracking of a robotic arm, or in [22], to improve the performance of anti-lock braking in vehicles like cars, utilizing this control theory resulted in useful and satisfactory outcome.

The central idea in this method is transform dynamics of nonlinear systems (total or partial) into the linear form, in a way that linear control methods can be applied. In this method, instead of making linear approximation of the system, nonlinearities of the systems are removed using an appropriate variable change. Feedback linearization methods can be considered methods for converting original models of system to simpler forms.

This controller has been used extensively in aerospace researches. In this field, [23] deals with designing an Omnicopter and after deriving nonlinear equations, has established an appropriate attitude control for system by using feedback linearized input-output method. Important point in this study is using quaternions for describing attitude of the system. Many of researches in space science deals with attitude control of spacecraft with or without considering effects of actuators [24, 25]. Reference [26] has studied attitude control of a rigid spacecraft without considering actuators. In that study, stabilization was implemented satisfactorily with this control theory.

Reaction wheel is one of the most precise actuators that has been used in space technology. This actuator can be used in various numbers and geometries, in such a way that with changing configuration or number of actuator, the effect on attitude control can be changed. Reference [27] has studied three-axis orthogonal, pyramid, and tetrahedron configurations and regarding Input power. Based on results obtained from that study, pyramid configuration is a suitable geometrical configuration for actuators.

In this study, at first, the kinematic equations have been derived based on quaternions and attitude dynamics of a spacecraft, which also in dynamic section, effect of pyramid actuator configuration and existence of disturbance torque gravity gradient have been considered. Thereafter, design of feedback linearization controller is implemented on obtained equations. Finally, results for control law performance is presented in figures by simulation.

II. KINEMATIC EQUATIONS

Attitude maneuver in a spacecraft can expressed as the rotation of a rigid body in a certain coordinate system. These angles, in fact, are indicative of the amount of the rotation around each body-axes. It is possible to describe attitude maneuver around three body axes, using a single rotation around a single axis, which are called Euler angle and Euler axis, respectively. We can describe these concepts using quaternions, so that quaternion components can be defined by Euler angle and Euler axis. Quaternion comprises one scalar part and one vector part as follow:

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$$q = q_1 i + q_2 j + q_3 k + q_4 \tag{1}$$

If we use α for Euler angle, and e=[e_1 e_2 e_3] for components of unit vector for Euler axis, relationship between quaternions and Euler parameters is as follow:

$$q_{1-3} = e\sin(\alpha/2),\ \ q_4 = \cos(\alpha/2) \tag{2}$$

In order to avoid singularities with using Euler angles for expressing the attitude transformation, we use quaternion parameters. By defining Q=[q_1 q_2 q_3 q_4]^T and the body angular velocity vector as ω=[ω_x ω_y ω_z]^T, the time derivative of the quaternion becomes [28]:

$$\frac{d}{dt}\mathrm{Q} = \frac{1}{2}[\Omega']\mathrm{Q} \tag{3}$$

And Ω′ defined as

$$[\Omega'] = \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \tag{4}$$

III. Dynamic Equations

This section presents a review of dynamic equation of motion for a spacecraft. The spacecraft is assumed as a perfectly rigid body. The basic attitude dynamic of spacecraft is written in the form [28]:

$$T = \dot{h}_I = \dot{h} + \omega \times h \tag{5}$$

In previous equation, T is the external torque vector and h is the total angular momentum of spacecraft body. The total angular momentum will be divided between the angular momentum of body and the angular momentum of the reaction wheels.

$$h = h_B + h_W \tag{6}$$

In (6) h_w is the angular momentum of reaction wheels, which can be defined as in [27]:

$$h_W = LI_W\omega_W \tag{7}$$

And by choosing I_S as spacecraft inertia matrix, h_B is:

$$h_B = I_S\omega \tag{8}$$

Where I_W=diag(I_{w1}, $I_{w2, \dots}$, I_{wn}) is diagonal matrix of inertia and $[L]_{3\times n}$ is distribution matrix for reaction wheels and the angular velocity vector of reaction wheels is ω_W=[ω_{w1} ω_{w2} … ω_{wn}]^T. Index n represents the number of wheels used in control subsystem. Then (4) can be rewritten as follows:

$$\dot{h}_B = -\omega\times(h_B + h_W) - \dot{h}_W + T \tag{9}$$

The reaction wheel actuators work on the principle of momentum exchange. Therefore, the angular momentum produced by this actuator is transferred to the spacecraft body with opposite sign.

$$\dot{h}_W = -T_C \tag{10}$$

The following equation can be derived, in order to describe the dynamics of a spacecraft:

$$\frac{d\omega}{dt} = I^{-1}\left[-\omega\times(I\omega + h_W) + T + T_C\right] \tag{11}$$

In this study we suppose that external torques are equal zero.

IV. Distribution Matrix Of Reaction Wheels

Although, using more than three wheels causes an increase in weight and costs, but in order to maintain the ability of control in case of malfunctioning of one wheel and also for exerting some optimizing constraints, it is more preferential than using three actuators.

Distribution matrix is unique for every geometrical configuration and has n column, according to n actuator. In this study, pyramid configuration has been used, which, according to fig. 1, each actuator is deviated from plane x_b-y_b by angle value β. Then, this structure is rotated around axis z_b by angle value θ, as in fig. 2. Relationship between reaction wheels torques and command torques is as follow [27]:

$$\begin{bmatrix} T_{cx} \\ T_{cy} \\ T_{cz} \end{bmatrix} = L_{3\times4}\begin{bmatrix} T_{w1} \\ T_{w2} \\ T_{w3} \\ T_{w4} \end{bmatrix} \tag{12}$$

By supposing β=35.264° and θ=45°, distribution matrix becomes:

$$L = \begin{bmatrix} \cos\beta\cos\theta & -\cos\beta\sin\theta & -\cos\beta\cos\theta & \cos\beta\sin\theta \\ \cos\beta\sin\theta & \cos\beta\cos\theta & -\cos\beta\sin\theta & -\cos\beta\cos\theta \\ \sin\beta & \sin\beta & \sin\beta & \sin\beta \end{bmatrix} = \begin{bmatrix} \sqrt{3}/3 & -\sqrt{3}/3 & -\sqrt{3}/3 & \sqrt{3}/3 \\ \sqrt{3}/3 & \sqrt{3}/3 & -\sqrt{3}/3 & -\sqrt{3}/3 \\ \sqrt{3}/3 & \sqrt{3}/3 & \sqrt{3}/3 & \sqrt{3}/3 \end{bmatrix} \tag{13}$$

In order to obtain each vector component T_{wi}, optimum criteria is defined by minimizing vector T_w=[T_{w1} T_{w2} T_{w3} T_{w4}]^T. Therefore, we consider following Hamiltonian [26]:

$$H = \sum_{i=1}^{4} T_{wi}^2 \tag{14}$$

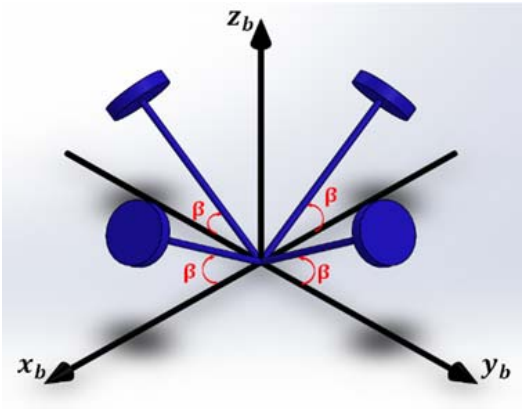


Fig. 1. The pyramid configuration of four reaction wheels

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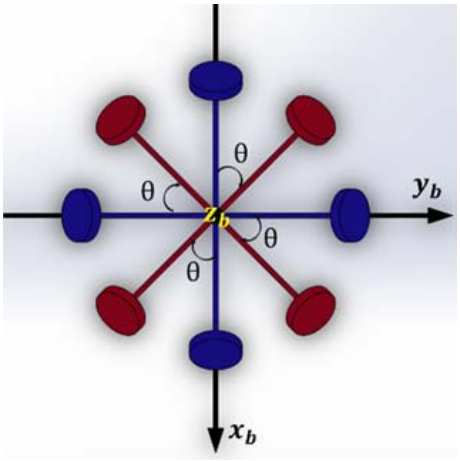


Fig. 2. Configuration rotation about z_b axis by an angle of ϴ

From (14) and matrix (13), we have:

$$\begin{aligned} T_{cx} &= \frac{\sqrt{3}}{3}(T_{w1} - T_{w2} - T_{w3} + T_{w4}) \\ T_{cy} &= \frac{\sqrt{3}}{3}(T_{w1} + T_{w2} - T_{w3} - T_{w4}) \\ T_{cz} &= \frac{\sqrt{3}}{3}(T_{w1} + T_{w2} + T_{w3} + T_{w4}) \end{aligned} \tag{15}$$

Now new equations are defined as (16):

$$\begin{aligned} g_1 &= \frac{\sqrt{3}}{3}(T_{w1} - T_{w2} - T_{w3} + T_{w4}) - T_{cx} \\ g_2 &= \frac{\sqrt{3}}{3}(T_{w1} + T_{w2} - T_{w3} - T_{w4}) - T_{cy} \\ g_3 &= \frac{\sqrt{3}}{3}(T_{w1} + T_{w2} + T_{w3} + T_{w4}) - T_{cz} \end{aligned} \tag{16}$$

Lagrangian is defined as:

$$L = H + \lambda_1 g_1 + \lambda_2 g_2 + \lambda_3 g_3 \tag{17}$$

And necessary conditions for minimizing H is,

$$\begin{aligned} \frac{\partial L}{\partial T_{w1}} &= 2T_{w1} + \frac{\sqrt{3}}{3}\lambda_1 + \frac{\sqrt{3}}{3}\lambda_2 + \frac{\sqrt{3}}{3}\lambda_3 \\ \frac{\partial L}{\partial T_{w2}} &= 2T_{w2} - \frac{\sqrt{3}}{3}\lambda_1 + \frac{\sqrt{3}}{3}\lambda_2 + \frac{\sqrt{3}}{3}\lambda_3 \\ \frac{\partial L}{\partial T_{w3}} &= 2T_{w3} - \frac{\sqrt{3}}{3}\lambda_1 - \frac{\sqrt{3}}{3}\lambda_2 + \frac{\sqrt{3}}{3}\lambda_3 \\ \frac{\partial L}{\partial T_{w4}} &= 2T_{w4} + \frac{\sqrt{3}}{3}\lambda_1 - \frac{\sqrt{3}}{3}\lambda_2 + \frac{\sqrt{3}}{3}\lambda_3 \end{aligned} \tag{18}$$

From (18), we can derive the last condition as,

$$\Delta T = T_{w1} - T_{w2} + T_{w3} - T_{w4} = 0 \tag{19}$$

Now, we can rewrite (12) by using this last condition.

$$\begin{bmatrix} T_{Cx} \\ T_{Cy} \\ T_{Cz} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} T_{w1} \\ T_{w2} \\ T_{w3} \\ T_{w4} \end{bmatrix} \tag{20}$$

This is a square matrix and therefore is reversible. So, for calculating actuator torques from command ones, we can use (21).

$$\begin{bmatrix} T_{w1} \\ T_{w2} \\ T_{w3} \\ T_{w4} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & 1 \\ -\frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & -1 \\ -\frac{3}{\sqrt{3}} & -\frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & 1 \\ \frac{3}{\sqrt{3}} & -\frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & -1 \end{bmatrix} \begin{bmatrix} T_{Cx} \\ T_{Cy} \\ T_{Cz} \\ 0 \end{bmatrix} = L_{WR} \begin{bmatrix} T_{Cx} \\ T_{Cy} \\ T_{Cz} \\ 0 \end{bmatrix} \tag{21}$$

V. Feedback Linearization Theory

This method wants to linearly describe the system using an appropriate variable change. The purpose is to find a control input for a nonlinear system, that nonlinearities are cancelled by using these inputs. Nonlinear equations of system and output are as follow:

$$\begin{aligned} \dot{x} &= f(x) + g(x)u \\ y &= h(x) \end{aligned} \tag{22}$$

According to (3) and (11), it is possible to define the state variables as $x=[q_1\ q_2\ q_3\ q_4\ \omega_x\ \omega_y\ \omega_z]^T$. Assuming this definition, matrices f(x) and g(x) are:

$$f(x) = \begin{bmatrix} \frac{1}{2}\Omega Q \\ I^{-1}\left[-\omega \times (I\omega + h_w) + T\right] \end{bmatrix} \tag{23}$$

$$g(x) = \left[g_1(x)\ g_2(x)\ g_3(x)\right] = \begin{bmatrix} 0_{4\times 3} \\ I^{-1} \end{bmatrix} \tag{24}$$

Also, $u=[u_1\ u_2\ u_3]^T=[T_{cx}\ T_{cy}\ T_{cz}]^T$ is input control of system and $y=[h_1(x)\ h_2(x)\ h_3(x)]^T=[q_1\ q_2\ q_3]^T$ is outputs of system. We can directly calculate the state variable q_4, using $q_1^2+q_2^2+q_3^2+q_4^2=1$.

Outputs of the system are not directly related to control inputs. In this control theory, as much the derivation form the output components continues, until the input appears.

By using Lie derivatives, input-output feedback linearization can convert a non-linear system to a linear one. Then, it is possible to use a linear control law for linearized system.

A. Feedback Llinearization Controller

Determination of output function from state variables of system has a direct relationship to creating internal dynamics and stability. According to presented model at [26], by considering state variable vector, the vector relative degree of the system (22) is $[r_1\ r_2\ r_3]^T= [2\ 2\ 2\ ]^T$.

According to [29], difference between total relative degree and dimensions of system means the existence of internal dynamics that has been studied in zero-dynamics section. This is in a case that main system has a degree of 7. It can be said that:

$$\begin{bmatrix} y_1^{(r_1)} \\ y_2^{(r_1)} \\ y_3^{(r_1)} \end{bmatrix} = D(x) + E(x)u \tag{25}$$

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Where D(x) and E(x) are,

$$D(x)=\begin{bmatrix} L_f^{r_1}h_1(x) \\ L_f^{r_2}h_2(x) \\ L_f^{r_3}h_3(x) \end{bmatrix} \tag{26}$$

$$E(x)=\begin{bmatrix} L_{g_1}L_f^{r_1-1}h_1(x) & \cdots & L_{g_3}L_f^{r_1-1}h_1(x) \\ \vdots & \ddots & \vdots \\ L_{g_1}L_f^{r_3-1}h_3(x) & \cdots & L_{g_3}L_f^{r_3-1}h_3(x) \end{bmatrix} \tag{27}$$

Lie derivatives used here have definition of:

$$L_f h_i(x)=\sum_{j=1}^{7}\frac{\partial h_i}{\partial x_i}f(x), \tag{28}$$

$$L_f^{r_i}h_i(x)=L_f\left(L_f^{r_i-1}h_i(x)\right)=\sum_{j=1}^{7}\frac{\partial L_f^{r_i-1}h_i}{\partial x_i}f(x), \tag{29}$$

$$L_{g_i}L_f^{r_i-1}h_i(x)=\sum_{j=1}^{7}\frac{\partial L_f^{r_i-1}h_i}{\partial x_i}g_i(x), i=1,2,3. \tag{30}$$

If det(E(x))≠0, this means that matrix E(x) is not singular, and feedback linearization is possible.

$$E(x)=\frac{1}{2}\begin{bmatrix} \frac{q_4}{I_{xx}} & \frac{-q_3}{I_{yy}} & \frac{q_4}{I_{zz}} \\ \frac{q_3}{I_{xx}} & \frac{q_4}{I_{yy}} & \frac{-q_1}{I_{zz}} \\ \frac{-q_2}{I_{xx}} & \frac{q_1}{I_{yy}} & \frac{q_4}{I_{zz}} \end{bmatrix} \tag{31}$$

$$det\left(E(x)\right)=\frac{q_4\left(q_1^2+q_2^2+q_3^2+q_4^2\right)}{8I_{xx}I_{yy}I_{zz}}=\frac{q_4}{8I_{xx}I_{yy}I_{zz}} \tag{32}$$

If $q_4\neq 0$, the matrix E(x) is non-singular. By defining ν as the new control input for linear system of

$$v=D(x)+E(x)u \tag{33}$$

Main control equation can be defined as,

$$u=E^{-1}(x)\left(v-D(x)\right) \tag{34}$$

B. Linearized System

Using feedback linearization method, non-linear system of (22) converts to a controllable linear input-output system in appropriate coordination. Variable of linear system can be defined as:

$$\begin{aligned} \xi_1&=h_1(x)=q_1 \quad & \xi_4&=L_f h_1(x)=\dot{q}_1 \\ \xi_2&=h_2(x)=q_2 \quad & \xi_5&=L_f h_2(x)=\dot{q}_2 \\ \xi_3&=h_3(x)=q_3 \quad & \xi_6&=L_f h_3(x)=\dot{q}_3 \end{aligned} \tag{35}$$

Linearized system in new coordination is presented in the following equations:

$$\begin{aligned} \dot{\xi}&=A\xi+Bv \\ y&=C\xi \end{aligned} \tag{36}$$

Where $\xi=[\xi_1\ \xi_2\ \xi_3\ \xi_4\ \xi_5\ \xi_6]^T$ is state variable matrix , and also we have:

$$\begin{aligned} A&=\begin{bmatrix} 0_{3\times3} & I_{3\times3} \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix} \\ B&=\begin{bmatrix} 0_{3\times3} \\ I_{3\times3} \end{bmatrix} \\ C&=\begin{bmatrix} I_{3\times3} & 0_{3\times3} \end{bmatrix} \end{aligned} \tag{37}$$

Now, for linear system of (36), using linear control laws, a controller can be designed. For this linear system, we consider input control of attitude error (38), in which K_1 and K_2 are coefficients matrices which are constant and positive.

$$v=-K_1(q-q_d)-K_2\dot{q} \tag{38}$$

In this equation q_d is the desired quaternion. For determining coefficients matrices K_1 and K_2, we use linear quadratic regulator method from optimal control theory.

C. Linear Quadratic Regulator for Linear System

One of the optimal control techniques that provides the best performance with respect to a criteria, is linear quadratic regulator method. The purpose of LQR is to design a state feedback controller *K* such that objective function *J* is minimized [30].

$$J=\int_0^{\infty}(x^TQx+u^TRu)dt \tag{39}$$

In the (39), *Q* and *R* are the weight matrices. To minimize the value of this cost function, the feedback control law is given by

$$u=-Kx \tag{40}$$

Where *K* is feedback coefficients matrix and is obtained by (41) and *P* is found by solving algebraic riccati equation in (42).

$$K=R^{-1}B^TP \tag{41}$$

$$A^TP+PA+Q-PBR^{-1}B^TP=0 \tag{42}$$

D. Zero Dynamics

As we mentioned earlier, difference between total relative degree and main dimensions of system equals one that implies the existence of an internal dynamic. We must consider that state variable q_4 is not related to output y. We must be sure that variable q_4 have a good behavior. This is the same aforementioned internal dynamics that its stability is investigated by applying zero-dynamics concept. For this purpose, parameter η is defined as zero-dynamics parameter that has to satisfy two important requirements [23]:

- $(\partial\eta/\partial x)g(x)=0$;
- The transform $z=T(x)=[\eta;\xi]$ must be a diffeomorphism.

The variable of zero-dynamics is chosen as $\eta=q_4$. So,

$$\frac{\partial\eta}{\partial x}g(x)=[1000000]\begin{bmatrix} 0_{4\times3} \\ I^1 \end{bmatrix}=0 \tag{43}$$

In order to investigate the diffeomorphism of z=T(x), the Jacobean matrix z must be generated, and its determinant must be determined:

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$$J(z)=\frac{\partial z}{\partial x}=\begin{bmatrix} I_{4\times4} & 0_{4\times3} \\ \Lambda(\omega) & sk(Q) \end{bmatrix} \tag{44}$$

$$\Lambda(\omega)=\begin{bmatrix} \frac{\omega_x}{2} & 0 & \frac{\omega_z}{2} & -\frac{\omega_y}{2} \\ \frac{\omega_y}{2} & -\frac{\omega_z}{2} & 0 & \frac{\omega_x}{2} \\ \frac{\omega_z}{2} & \frac{\omega_y}{2} & -\frac{\omega_x}{2} & 0 \end{bmatrix} \;\&\; sk(Q)=\begin{bmatrix} \frac{q_4}{2} & -\frac{q_3}{2} & \frac{q_2}{2} \\ \frac{q_3}{2} & \frac{q_4}{2} & -\frac{q_1}{2} \\ -\frac{q_2}{2} & \frac{q_1}{2} & \frac{q_4}{2} \end{bmatrix} \tag{45}$$

$$\det\left(J(x)\right)=\frac{q_4}{8}\left(q_1^2+q_2^2+q_3^2+q_4^2\right)=\frac{q_4}{8} \tag{46}$$

If $q_4 \neq 0$ the zero dynamics satisfies the two requirements and the Jacobian matrix is nonsingular. Then using the feedback linearization controller and under this condition, the system will be stable. Also the mapping z will be diffeomorphism.

VI. Simulation

In this section, in order to study the performance of designed attitude control system, an attitude maneuver has been considered, that is a maneuver with large command angles. This maneuver starts from initial position of $[\phi_0\ \theta_0\ \psi_0]= [0\ 0\ 0\ ]$ towards the final position of $[\phi_f\ \theta_f\ \psi_f]=[10\ -30\ 40]$, based on degree.

Spacecraft weight is 60 Kg, Inertial matrix and inertial of wheels are diag(4,4,3) and 0.0005 Kgm^2 respectively, Orbital radius is 600 Km. Weight matrices for minimizing criteria function (39) has been chosen, based on results from simulations.

$$Q_{LQR}=0.1\times I_6 \tag{47}$$

$$R_{LQR}=100\times I_3 \tag{48}$$

Selection of these weight matrices causes the controlling proportional-derivative coefficient matrices to appear like following:

$$K_1=0.0316\times I_3 \tag{49}$$

$$K_2=0.2535\times I_3 \tag{50}$$

Simulation of the attitude maneuver has been carried out for a time range of 60 seconds, and with the assumption that the reaction wheels are without speed in beginning of the maneuver. This simulation carried out by MATLAB and results are presented in fig. 3 to fig. 8.

Consider fig. 4, the large angle maneuver begins from initial conditions, without perceptible overshoot, with desirable and non-oscillatory behavior reaches to commanded angles. Fig. 3 shows quaternion parameters along maneuver.

As it is seen in fig. 5, spacecraft body, from zero speed begins to rotate around three body axes and angular velocity of spacecraft changes. But these velocity changes of body, completely disappear as maneuver finishes, and the system becomes static again in its new attitude.

According to (38), control input tries to eliminate the attitude error, and this error is clearly recognizable in time 0 in fig. 7. In Fig. 8, torque contribution of every actuator, to reach desirable torques around body-axes is given.

The torque of each actuator generates due to angular acceleration of wheels. In fig. 6, necessary change of angular velocity for maneuver time duration is presented.

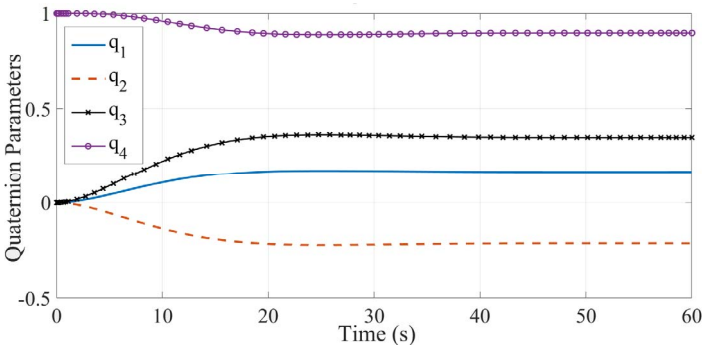


Fig. 3. Quaternion Parameters

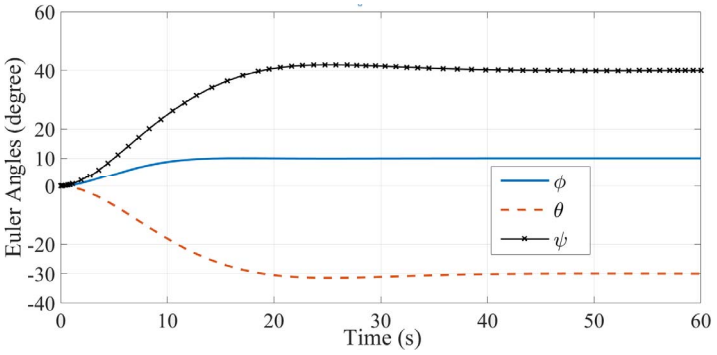


Fig. 4. Euler Angles

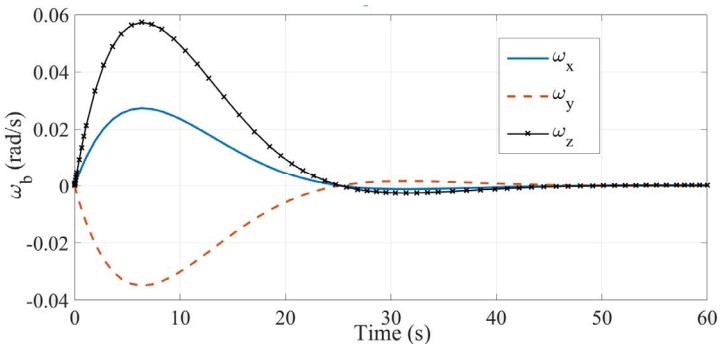


Fig. 5. Spacecraft Angular Velocities

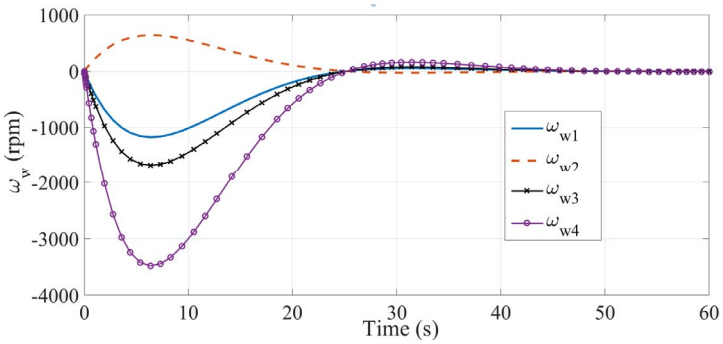


Fig. 6. Angular Velocities of Reaction Wheels

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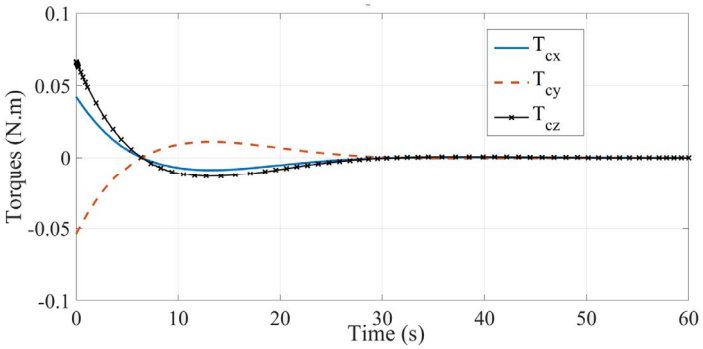


Fig. 7. Control Torques

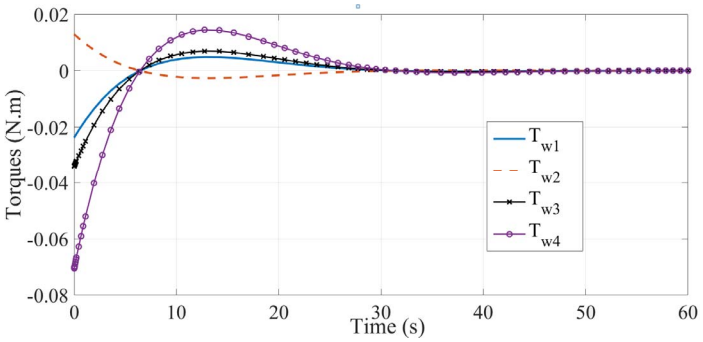


Fig. 8. Reaction Wheels Torques

VII. Conclusion

The spacecraft model that is used in this study includes reaction wheel configuration. In order to control this nonlinear system, a nonlinear control theory is considered. Feedback linearization is an approach to nonlinear control design that is based on the idea of transforming nonlinear dynamics into a linear form by using state feedback, so that linear control techniques can be applied. Optimal quadratic regulator as a linear control has selected here. Since the kinematic equations are expressed by quaternion parameters, these state variables have chosen as the output functions. The nonlinearities in nonlinear system are cancelled through control input. Finally, based on the results it can be stated that input-output linearization method can be used for large angle maneuvers of a rigid body spacecraft by reaction wheel actuators.

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